



STEALTH BOMBER

The Northrop B-2 is a flying-wing design that uses angular faceting to reduce its radar signature. The radar-absorbent coating is so sensitive that the bombers must be kept in spotlessly clean hangars at controlled humidity levels.

The terrorist attacks in America on September 11, 2001 set off a new chain of events in which air power took center stage. The array of carrier-borne and land-based aircraft that the US brought to bear on Afghanistan in the “war against terrorism” proved effective, once more confounding critics who had suggested that only a considerable ground force could achieve America’s objectives. In contrast with the experience of the Soviet Union in Afghanistan in the 1980s, the US was able to depose the Taliban government and destroy al-Qaeda bases with minimal involvement of ground forces. No resistance seemed possible to a military force apparently capable of pinpointing targets

however mobile or well hidden they might be, and able to call upon anything from cruise missiles to unpiloted aircraft, fixed-wing gunships, and B-52 bombers to carry out the work of destruction.

Success and limitations

Critics of air power of course pointed out that these were exceptionally one-sided conflicts. NATO’s Kosovo campaign faced a Serbian regime that was politically weak and had limited will to resist. In Afghanistan, local opposition forced could be mustered to take over such political power as existed. When the United States and Britain invaded Iraq in 2003, there were pessimistic predictions that this time it would be different. Ground forces, committed in strength, would have to slog it out in costly battle with Saddam Hussein’s Republican Guard. Yet once more air power proved its absolute worth. Media attention focused on the “shock and awe” spectacle of the destruction of Saddam’s palaces and government offices in Baghdad, but the great majority of aerial firepower was directed at Iraqi troops and armor. Since Iraq’s air defense system had been systematically dismantled by air attacks in the run-up to the invasion, American and British aircraft operated at will. Every movement of Iraqi forces was tracked by Allied reconnaissance, including Global Hawk UAVs, and air strikes were directed accurately on to targets even in adverse weather conditions. In effect, the function of Allied ground forces was to draw Iraqi forces into action so that aircraft could destroy them.

Air power brought swift victory in the initial invasion of Iraq, but the aftermath showed its limitations. In the face of a rising tide of insurgency and terrorism, the effectiveness of air power was

restricted by the difficulty of identifying targets and by political sensitivity about civilian casualties caused by air attack. However accurate the weaponry, there were always going to be mistakes due to faulty intelligence, equipment malfunction, or simple human error. Unleashed in total war, American air power could have hit any target in Iraq that was deemed worth destroying. But it could not stop suicide bombers.

Next generation

The time-lag involved in development of new aircraft meant that in the early 21st century the world’s major air forces were still essentially designed to fight the Cold War. The B-2 stealth bomber, for example, was meant to penetrate Soviet air defenses carrying a nuclear payload. The Lockheed Martin F-22 Raptor, which entered service with the US Air Force in 2005, was originally conceived as a fighter that would trounce its Soviet equivalents in air-to-air combat. The future relevance of such aircraft was bound to be called into question, especially as they were immensely expensive—the B-2 program reportedly came in at around \$45 billion, although only 21 of the bombers were built.

Faced with the diffuse threat of global terrorism, the United States looked to a future in which the long reach of air power could be used to strike a vehicle or building anywhere around the world, identified as containing enemies of America. Satellites and unpiloted aircraft were central to this vision of global surveillance and precisely targeted vengeance. Nevertheless, more conventional piloted fighters and bombers looked certain to continue to be produced in large numbers in the immediate future. In the 21st century, whatever its actual defense needs, no country with any pretensions to international status could afford to miss out on a truly modern air force, possession of which had become a prime symbol of technological superiority.

UNMANNED AERIAL VEHICLES

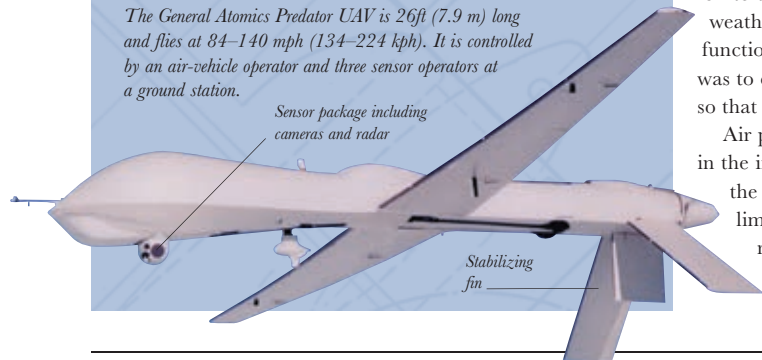
THE USEFULNESS OF DRONES—variously known as remotely piloted vehicles (RPVs) or unmanned aerial vehicles (UAVs)—has been apparent since the early 1980s. Although they can hardly be regarded as disposable, UAVs can carry out reconnaissance missions at a risk level that would be considered unacceptable for manned aircraft. “Flown” by a technical team on the ground, they transmit back television images or other data—for example, from radar and infrared sensors—often staying over enemy positions for many hours. During the US intervention in Afghanistan in 2001–02, RQ-1 Predator UAVs were used for the first time in an attack role, firing Hellfire air-to-ground missiles. Also introduced in Afghanistan were high-altitude RQ-4A Global Hawk UAVs, equipped for all-weather reconnaissance.

RQ-1 PREDATOR

The General Atomics Predator UAV is 26ft (7.9 m) long and flies at 84–140 mph (134–224 kph). It is controlled by an air-vehicle operator and three sensor operators at a ground station.

Sensor package including cameras and radar

Stabilizing fin



RAPTOR AND FALCON

An F-22 Raptor, a fighter designed for the 21st century, flies on the left of an F-16 Falcon. The differences between the two are subtle, although they add up to a major advance. The F-22 has stealth features including the use of composite materials; it can cruise at around Mach 1.4, rather than going supersonic only in short bursts; it has vectored thrust to increase its agility; and has astonishingly sophisticated electronics and computer systems.